Supporting Information

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S1 Fig. Whole-heart cohort reconstructions.

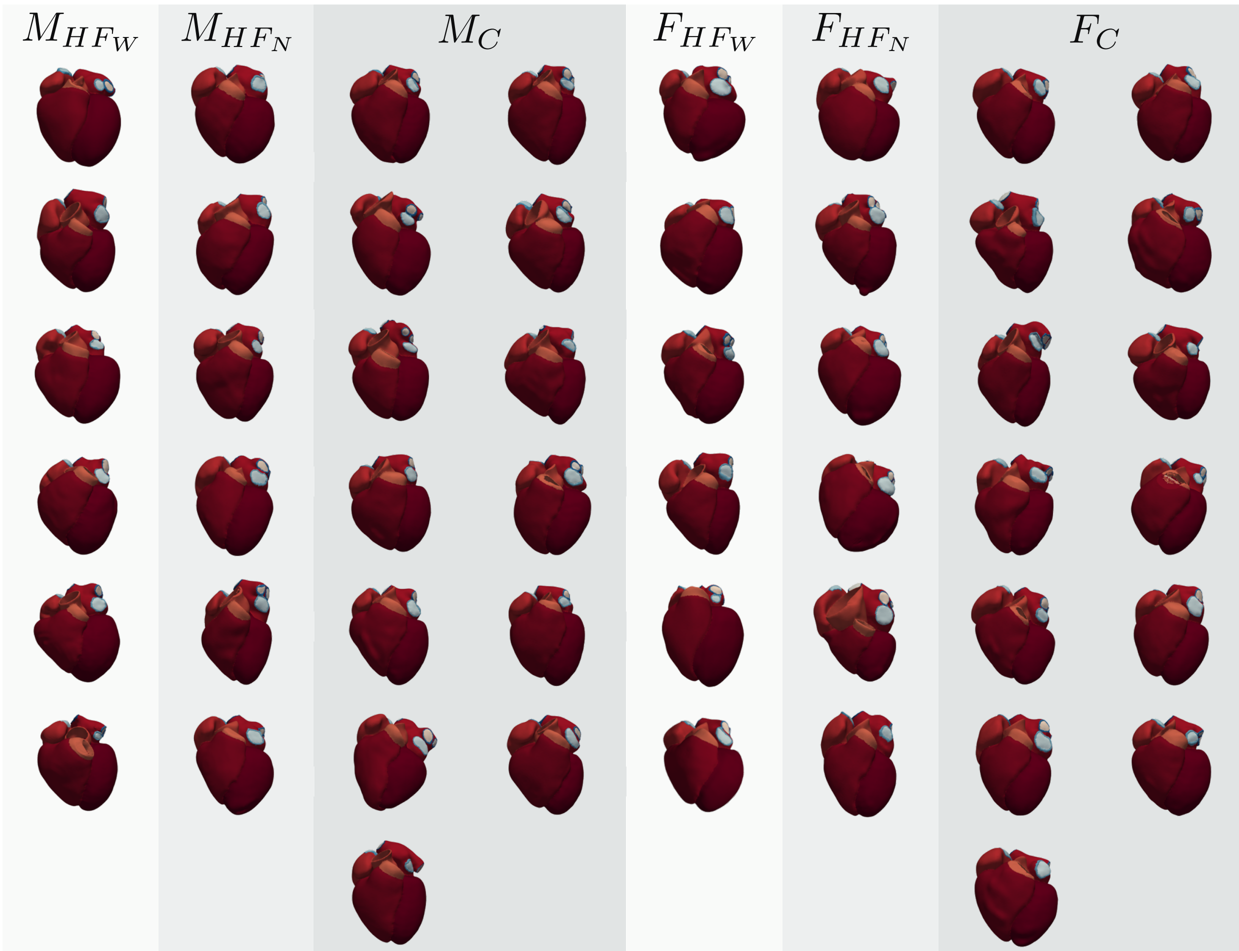

Fig S1. Three-dimensional whole-heart reconstructions from the virtual cohort, grouped and labelled by sex and condition. Each colour corresponds to a different structure in the mesh. Each model was derived from a patient's CT scan and includes fibre orientations, illustrating anatomical variability across sex and heart-failure subtypes. Abbreviations: M_{HF_N}: Male, HF narrow-QRS; M_{HF_W}: Male, HF wide-QRS; M_C: Male, Control; F_{HF_N}: Female, HF narrow-QRS; F_{HF_W}: Female, HF wide-QRS; F_C: Female, Control.

S2 Fig. TAT-QRS correlation, stratified by condition.

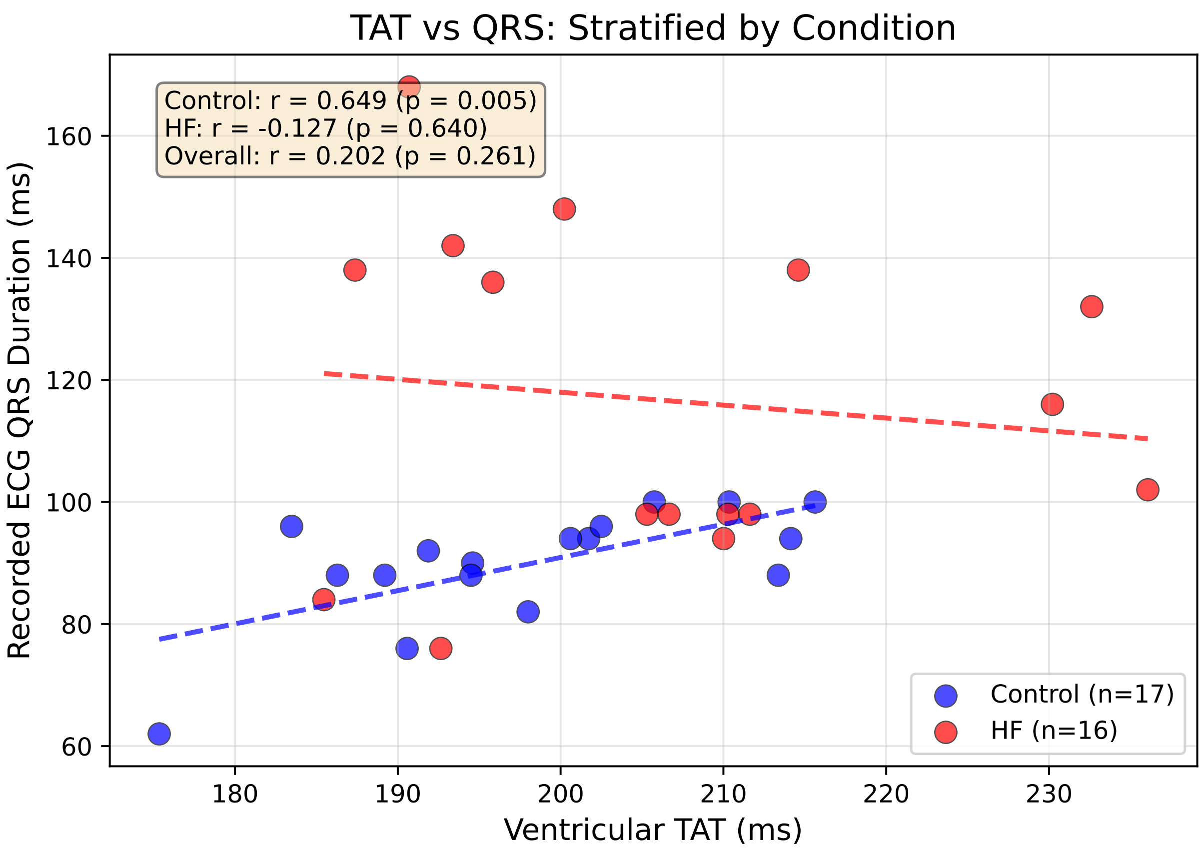

Fig S2. Correlation between simulated ventricular total activation time (TAT_V) and recorded ECG QRS duration, stratified by condition. In C hearts (blue, $n = 17$), TAT_V showed a strong positive correlation with QRS duration (Pearson $r = 0.649$, $p = 0.005$), indicating that anatomical variability alone can predict conduction time in healthy myocardium when using uniform electrophysiological parameters. In HF patients (red, $n = 16$), no correlation was observed ($r = -0.127$, $p = 0.64$), reflecting that pathological conduction delays (fibrosis, scar, conduction blocks) are not captured by anatomy alone and would require patient-specific tissue property calibration.

S1 Table. Geometric characteristics by sex and heart failure status.

S3 Fig. Geometry-output correlation matrix.

S2 Table. CGAL meshing parameters.

Mesh density is controlled via parameters of the Computational Geometry Algorithms Library (CGAL) [1], specified globally for the entire heart geometry. Regional refinement was not applied; a single set of parameters governs element density and surface fidelity across all anatomical structures.

S3 Table. Mesh label specification.

The post-processing stage of the CEMRG Heartbuilder pipeline increases the initial 10-label segmentation to 37 labels (Fig~~Figure~~[1] 2c), following the standard introduced by [2,3]. Labels encode anatomical structures

[1] **[PLOS 1]:** PLOS: Fig (no period)

Table S1. Geometric characteristics stratified by sex and heart failure status. Mean ± SD reported for age (years), chamber volumes (mL), and myocardial wall volume (cm^3). HF patients showed increased chamber volumes across all chambers, particularly pronounced in males. Myocardial wall volume scaled with anatomical size, reflecting whole-heart structural differences across groups.

Group	n	**Age (yr)**	Vol_{LV} **(mL)**	Vol_{RV} **(mL)**	Vol_{LA} **(mL)**	Vol_{RA} **(mL)**	**Wall Vol. (cm^3)**
Male C	13	47.2 ± 12.0	179.6 ± 36.3	188.7 ± 25.7	93.9 ± 27.9	100.7 ± 16.9	319.0 ± 43.7
Male HF	12	60.1 ± 16.1	289.1 ± 88.1	227.6 ± 60.2	140.5 ± 58.9	138.2 ± 67.7	396.0 ± 75.2
Female C	13	49.5 ± 9.8	139.7 ± 32.2	139.7 ± 34.8	79.6 ± 19.4	86.3 ± 32.3	262.6 ± 36.5
Female HF	12	60.1 ± 9.0	183.0 ± 60.0	136.7 ± 30.8	104.9 ± 57.4	100.0 ± 47.3	297.3 ± 50.0

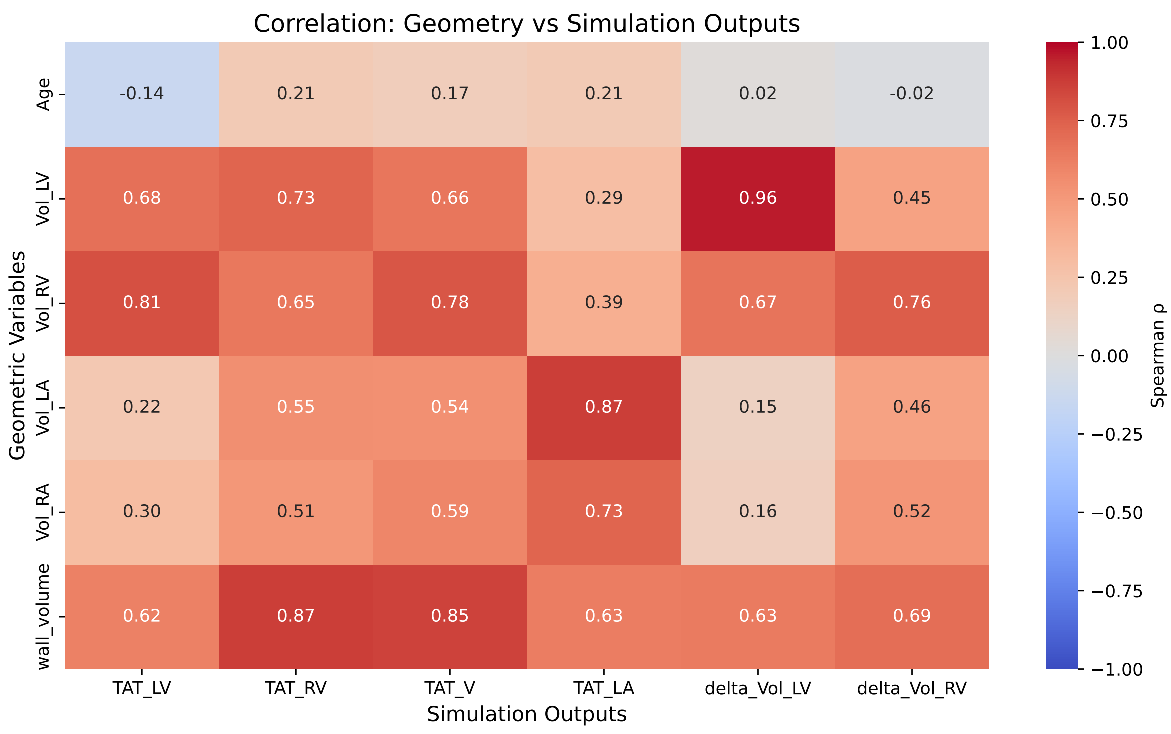

Fig S3. Correlation matrix between geometric variables and simulation outputs. Spearman correlation coefficients (ρ) shown for relationships between geometric metrics (chamber volumes, myocardial wall volume, age) and simulation outputs (activation times, volume changes). Strong positive correlations (red) indicate that geometric variables predict functional outcomes under standardised parameters. All correlations shown are statistically significant ($p < 0.05$).

Table S2. CGAL meshing parameters used for volumetric mesh generation. Parameters are specified globally for the entire heart geometry. `cell_size` and `facet_size` set the target element edge length; `facet_distance` controls the surface approximation error (trade-off between surface fidelity and smoothness). All 50 meshes were generated with identical parameter values.

Parameter	**Value**	**Effect on mesh**
`cell_size`	≈ 0.5 mm	Maximum target edge length of volumetric tetrahedral elements; controls overall mesh density.
`facet_size`	≈ 0.5 mm	Maximum target size of surface facets; controls surface mesh resolution and matches `cell_size` for uniform density.
`facet_distance`	4	Maximum Hausdorff distance between the input surface and the Delaunay surface; Laplacian smoothing is applied to balance surface fidelity against smoothness.
`cell_radius_edge_ratio`	2.0	Upper bound on the ratio of circumradius to shortest edge; controls element shape quality.

required for assigning tissue-specific electrophysiology and mechanics model parameters during simulation setup.

S1 Text. Mechanics simulation boundary conditions.

As boundary conditions, the pericardial sac was modelled using normal springs with varying stiffness and omni-directional springs were applied at the superior vena cava and the right pulmonary veins [4]. All the tissues were modelled as incompressible with an incompressibility value of $\kappa = 1000$ kPa. The bulk stiffness of left ventricle, right ventricle and atria were set to $a_{LV} = 1.7$ kPa, $a_{RV} = 2.55$ kPa, and $a_{\text{atria}} = 3.4$ kPa, respectively. The anisotropy factors in the ventricles in the fibre, cross-fibre and cross-sheet directions were set to $b_f = 8$, $b_{fs} = 4$, and $b_t = 3$, respectively; while in the atria were set to $b_f = 16$, $b_{fs} = 8$, and $b_t = 6$. The valves, aorta, pulmonary artery and vein rings had a stiffness value of $c_{\text{valves}} = 1000$ kPa, $c_{\text{aorta}} = 26.66$ kPa, $c_{\text{PA}} = 3.7$ kPa, and $c_{\text{rings}} = 7.45$ kPa, respectively. The maximum stiffness at the pericardium was set to 0.005 kPa/μm and to 0.001 kPa/μm in the veins. The pressure was increased linearly from 0 to 1 mmHg in increments of 0.02 mmHg/ms.

S4 Fig. Mesh quality assessment.

Mesh quality was assessed using the volume-based tetrahedral distortion metric (`tet_qmetric_volume`) implemented in meshtool, directly related to the Jacobian determinant of the geometric map. The metric ranges from 0 (perfect tetrahedron) to 1 (fully degenerate element). No inverted elements were detected in any of the 50 meshes, confirming numerical suitability for finite element analysis.

References

1. CGAL 4.11 - Manual: Package Overview;. Available from: `https://doc.cgal.org/4.11/Manual/packages.html`.

2. Strocchi M, Augustin CM, Gsell MAF, Karabelas E, Neic A, Gillette K, et al. A publicly available virtual cohort of four-chamber heart meshes for cardiac electro-mechanics simulations. PLOS ONE. 2020;15(6):e0235145. Publisher: Public Library of Science. doi:10.1371/journal.pone.0235145.

3. Strocchi M, Gsell MAF, Augustin CM, Razeghi O, Roney CH, Prassl AJ, et al. Simulating ventricular systolic motion in a four-chamber heart model with spatially varying robin boundary conditions to model the effect of the pericardium. Journal of Biomechanics. 2020 Mar;101:109645. doi:10.1016/j.jbiomech.2020.109645.

Table S3. Complete list of 37 anatomical labels. Labels are assigned during the post-processing stage of the segmentation pipeline. Structure types: BP = blood pool, Myo = myocardium, VP = valve plane, Ring = vein/artery ring.

Label ID	Structure name	Chamber	Type
1	Left ventricular blood pool	LV	BP
2	Left ventricular myocardium	LV	Myo
3	Right ventricular blood pool	RV	BP
4	Left atrial blood pool	LA	BP
5	Right atrial blood pool	RA	BP
6	Aortic blood pool	—	BP
7	Pulmonary artery blood pool	—	BP
8	Left superior pulmonary vein	LA	BP
9	Left inferior pulmonary vein	LA	BP
10	Right superior pulmonary vein	RA	BP
11	Right inferior pulmonary vein	RA	BP
12	Left atrial appendage	LA	BP
13	Superior vena cava	RA	BP
14	Inferior vena cava	RA	BP
103	Right ventricular myocardium	RV	Myo
104	Left atrial myocardium	LA	Myo
105	Right atrial myocardium	RA	Myo
106	Aortic wall	—	Myo
107	Pulmonary artery wall	—	Myo
201	Mitral valve plane	LV/LA	VP
202	Tricuspid valve plane	RV/RA	VP
203	Aortic valve plane	LV	VP
204	Pulmonary valve plane	RV	VP
205	Left superior pulmonary vein plane	LA	VP
206	Left inferior pulmonary vein plane	LA	VP
207	Right superior pulmonary vein plane	LA	VP
208	Right inferior pulmonary vein plane	LA	VP
209	Left atrial appendage plane	LA	VP
210	Superior vena cava plane	RA	VP
211	Inferior vena cava plane	RA	VP
221	Left superior pulmonary vein ring	LA	Ring
222	Left inferior pulmonary vein ring	LA	Ring
223	Right superior pulmonary vein ring	LA	Ring
224	Right inferior pulmonary vein ring	LA	Ring
225	Left atrial appendage vein ring	LA	Ring
226	Superior vena cava vein ring	RA	Ring
227	Inferior vena cava vein ring	RA	Ring

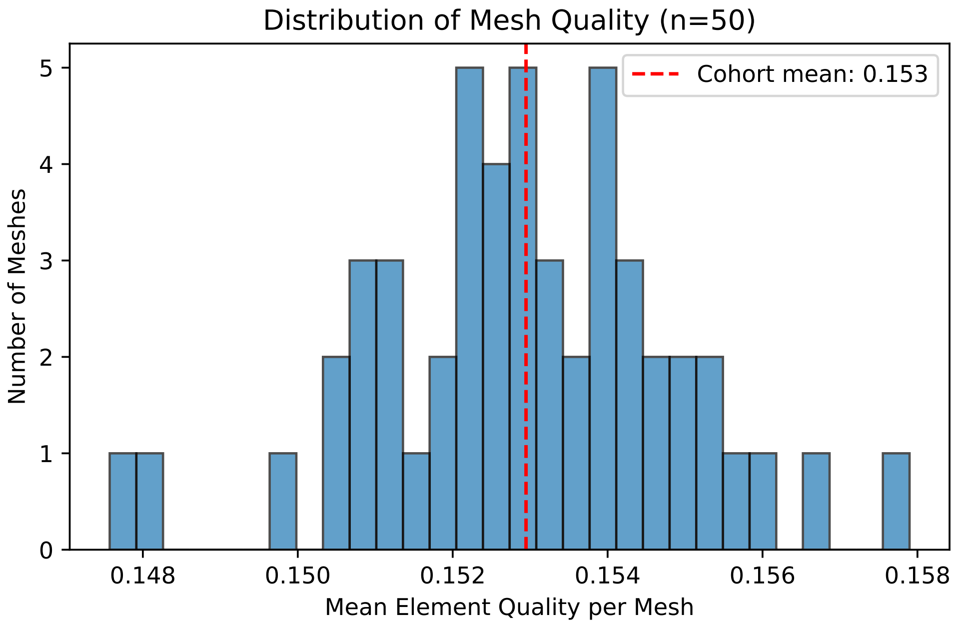

Fig S4. Distribution of mean element quality across the 50-heart cohort. Each point represents one mesh. The `tet_qmetric_volume` metric ranges from 0 (perfect tetrahedron) to 1 (degenerate); lower values indicate better quality. Cohort-wide mean was 0.153 ± 0.100. No inverted elements (quality > 0.99) were found in any mesh.

4. Strocchi M, Longobardi S, Augustin CM, Gsell MAF, Petras A, Rinaldi CA, et al. Cell to whole organ global sensitivity analysis on a four-chamber heart electromechanics model using Gaussian processes emulators. PLOS Computational Biology. 2023 Jun;19(6):e1011257. Publisher: Public Library of Science. doi:10.1371/journal.pcbi.1011257.